Regression Algorithms for Learning & Its Application on Crypto Currency Markets

Retrospective Report

Syed Maesum Hussain Zaidi

Undergraduate Final Year Project

Supervised By : Xu Feng

Department of Computer Science

Royal Holloway, University of London

1. Timeline Comparison (Planned vs Actual Execution)

At the start of the project, a detailed timeline was produced for the project plan outlining the expected progression of work across Term 1. The plan proposed a structured approach, beginning with theoretical foundations, followed by proof-of-concept development, literature review, and finally the application of regression algorithms to real-world cryptocurrency data.

In practice, the overall structure of the timeline was largely adhered to, although some adjustments were required as the project progressed. During Weeks 3-4, the planned focus on studying the mathematical foundations of linear, ridge, and lasso regression was achieved as expected, forming a solid theoretical base. Weeks 5-6 were originally intended to include both the completion of the first report and the development of a proof-of-concept program. While the first report was completed on schedule, the proof-of-concept implementation was delayed due to competing workload from other modules. This deviation was identified and documented in the project diary.

The delayed proof-of-concept was successfully completed in Week 7, allowing the project to realign with the original plan. The literature review and second report, scheduled for Weeks 7–8, were completed within the planned timeframe. Similarly, the third report, which aimed to connect theoretical foundations with the proof-of-concept, was completed in Week 9 as intended.

The most significant practical work occurred during Weeks 10–11, where the plan called for loading and preprocessing real BTC-USD data and implementing ridge regression. In practice, this stage exceeded the original scope. Not only were OLS and ridge regression implemented, but additional comparative analysis was performed, and model evaluation techniques such as walk-forward validation were introduced. In Week 12, the original plan was further extended with the addition of Lasso regression and automatic alpha tuning, replacing the earlier scikit-learn comparison.

Overall, while minor delays occurred in the early stages, the project successfully recovered and ultimately progressed beyond the originally planned technical depth.

2. Reflection on Term 1 Progress

Reflecting on Term 1, several aspects of the project progressed particularly well. The initial emphasis on theoretical understanding proved highly beneficial, as it enabled the later implementation of regression algorithms without reliance on built in implementations from libraries such as sci-kit learn. Writing the foundational reports before extensive coding helped clarify the mathematical assumptions and limitations of each model, which directly informed later design decisions.

One key challenge encountered was time management during the early stages of implementation. The initial underestimation of workload from other modules resulted in a short delay to the proof-of-concept program. However, this experience highlighted the importance of maintaining visible progress on version control platforms, even during non-coding phases. Following feedback from the supervisor, regular diary updates and Git commits were maintained, improving project tracking and accountability.

Another positive outcome was the decision to move from a simple proof-of-concept to a more realistic walk-forward evaluation strategy. Although not explicitly stated in the original plan, this adjustment significantly improved the robustness and credibility of the results. This reflected an increased understanding of time-series modelling and the importance of avoiding look-ahead bias in financial prediction tasks.

Overall, Term 1 has demonstrated steady academic and technical progression. The challenges faced were resolved through adaptation rather than compromise, and the lessons learned regarding planning, validation, and incremental development will be carried forward into Term 2.

3. Term 2 Plans

Term 2 will focus on extending the existing implementation into a more complete and user-friendly software system.

In week 2-3, the first milestone will involve refactoring the current procedural code into a fully object-oriented design, aligning the implementation with standard software engineering principles.

Following this, in week 4-6, a graphical user interface will be developed using a Python framework such as Tkinter or PyQt. This interface will allow users to select parameters, visualise model outputs, and interact with prediction results without modifying code directly.

In week 7-8, additional regression algorithms, such as nearest neighbours and neural networks, will then be implemented and evaluated against the existing OLS, ridge, and lasso models.

Week 9-12, The final phase of Term 2 will involve systematic experimentation, comparison of model performance, and comprehensive analysis of results. The final report will consolidate theoretical foundations, software engineering decisions, experimental findings, and critical discussion, thus resulting in a complete and coherent project deliverable.

4. Risk Assessment and Mitigation Review

Several risks identified in the original project plan were encountered during Term 1, most notably time management and computational complexity. The risk of falling behind schedule materialised briefly during the proof-of-concept stage, but mitigation strategies such as timeline adjustment and task prioritisation proved effective.

Concerns regarding overfitting and evaluation bias were addressed through the adoption of walk-forward validation and multiple performance metrics, including mean squared error and directional accuracy. Hence, these risks were mitigated.

Looking ahead, remaining risks primarily concern scalability, user interface complexity, and computational cost when introducing more advanced models. These will be mitigated through modular design, careful algorithm selection, and controlled experimentation during Term 2.

5. Conclusion

In conclusion, the retrospective analysis demonstrates that the project has progressed in a structured, reflective, and academically sound manner. While minor deviations from the original timeline occurred, these were managed effectively and ultimately strengthened the project's technical depth. The experience gained during Term 1 has provided a strong foundation for the successful completion of the final project in Term 2.